Continuous Optimization in Data Science Assignment 2

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Total points: 63pt.

Q1. Logistic Regression

1. We consider a dataset $\{(y_i \in \{-1,1\}, \mathbf{x}_i \in \mathbb{R}^n)\}$. We want to find a relation between y_i and $\mathbf{x}_i$, but y_i is binary. When y_i takes discrete values, we call such a problem "classification", because samples with a similar value y_i can be said to be from the same class. The example that we consider is related to breast cancer, as the last assignment. $\mathbf{x}$ is a list of activity level of 10 different genes and $y = -1$ means no cancer is detected, while $y = 1$ means cancer is detected. For a bianry variable $Y \in \{-1, 1\}$, we define its odds as the following ratio:

$$\text{odds}(Y) = \frac{\Pr[Y = 1]}{\Pr[Y = -1]} \tag{1}$$

   In logistic regression, we assume

$$\text{odds}(Y \mid \mathbf{x}) = e^{\boldsymbol{\theta}^T \mathbf{x} + \mu} \tag{2}$$

   where $\boldsymbol{\theta}, \mu$ are the model parameters. Write the negative log likelihood function for the dataset. (Hint: First, you need to calculate $\Pr[Y = \pm 1 \mid \mathbf{x}]$ from the odds) (5p).

2. The maximum likelihood problem that you obtain above is convex and hence can be efficiently solved. You can use its implementation in the sklearn library (look at cancer.py for more instructions). When loading the data, note that variable 1 is much bigger than the others. Apply logistic regression to the first 500 samples and learn $\boldsymbol{\theta}, \mu$. Consider two cases: In the first one, take variable 1 as is. In the second one, divide it by 10^6 in every sample. For both cases, calculate the predicted odds for the remaining 183 samples (by the learned model) as test samples. For every test sample with odds larger than 1, we predict that $y = 1$ and otherwise $y = 0$. Compare it to the actual labels and count how many predictions are correct. This is known as accuracy. Write a short report about the result (10p):

   (a) Does the second case (i.e. preprocessing the data) give you a different solution? Why do you think?

   (b) What can we learn from $\boldsymbol{\theta}$? is there any particular gene that has a bigger effect on cancer?

   (c) How well is the accuracy? Discuss what is the reason and what are the practical considerations for using the learned model.

3. The above is an example of a more general method of *linear classification*. Take a dataset $\{y_i \in \{-1,1\}, \mathbf{x}_i \in \mathbb{R}^m\}$. In linear classification, we ideally look for variables $\boldsymbol{\theta} \in \mathbb{R}^m, \mu \in \mathbb{R}$ such that for $y_i = +1$, we have $\boldsymbol{\theta}^T \mathbf{x}_i + \mu < 0$ and for $y_i = -1$, we have $\boldsymbol{\theta}^T \mathbf{x}_i + \mu > 0$. In short, we want to have $y_i(\boldsymbol{\theta}^T \mathbf{x}_i + \mu) < 0$. If we can do so, we say that the dataset is linearly separable and $\boldsymbol{\theta} \in \mathbb{R}^m, \mu \in \mathbb{R}$ are a separating hyperplane. Otherwise, we try to find the best approximation by the following ERM framework:

$$\min_{(\boldsymbol{\theta}, \mu)} \sum_{i=1}^{n} \ell\left(y_i(\boldsymbol{\theta}^T \mathbf{x}_i + \mu)\right) + \frac{\alpha}{2} \|\boldsymbol{\theta}\|_2^2 \tag{3}$$

where $\alpha \geq 0$ is a regularization parameter and $\ell(.)$ is a loss function that returns a small value for negative inputs and a large value for positive values. There are two common examples. The logistic loss function is given by

$$\ell(z) = \log\left(1 + e^{z}\right) \tag{4}$$

and leads to the logistic regression problem. Another choice of $\ell(.)$ is hinge loss (a.k.a rectified linear unit, ReLU):

$$\ell(z) = \begin{cases} 0 & z < 0 \\ z & z \geq 0 \end{cases} \tag{5}$$

This choice leads to support vector machine (SVM). A trick that is usually used in this setup is to rewrite (3) as follows:

$$\begin{aligned} &\min_{(\boldsymbol{\theta},\mu),\{z_i \in \mathbb{R}\}} \sum_{i=1}^{n} \ell(z_i) + \tfrac{\alpha}{2}\|\boldsymbol{\theta}\|_2^2 \\ &\text{s.t.} \\ &z_i = y_i(\boldsymbol{\theta}^T \mathbf{x}_i + \mu) \end{aligned} \tag{6}$$

(a) Write the Lagrangian function $L\left((\boldsymbol{\theta}, \mu), \{z_i \in \mathbb{R}\}, \boldsymbol{\lambda}\right)$ where $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \ldots, \lambda_n)$ is the vector of dual variables. Show that if ℓ is convex, then the problem in (6) satisfies Slater's condition (5p).

(b) Write the Lagrangian min-max optimization and by interchanging min and max show that for convex ℓ the dual optimization can be written as:

$$\begin{aligned} &\min_{\{\lambda_i \in \mathbb{R}\}} \sum_{i=1}^{n} \ell^*(\lambda_i) + \tfrac{1}{2\alpha} \left\| \sum_{i=1}^{n} y_i \mathbf{x}_i \lambda_i \right\|_2^2 \\ &\text{s.t.} \\ &\sum_{i=1}^{n} \lambda_i y_i = \mathbf{0} \end{aligned} \tag{7}$$

where ℓ^* is the convex conjugate of $\ell(.)$ (8p).

(c) Calculate the convex conjugate functions of

i. The logistic loss (6p)

ii. The hinge loss (6p)

Plot the functions (e.g. by Python matplotlib).

4. (15p) We would like to empirically calculate and compare the risk function of SVM and logistic regression. For this, write a code that generates n $d-$dimensional samples:

$$\mathbf{x}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{8}$$

and their binary labels y_i such that

$$\text{Odds}(y_i) = \frac{\Pr[y_i = +1]}{\Pr[y_i = -1]} = e^{\beta t_i}, \quad t_i = \boldsymbol{\theta}_0^T \mathbf{x}_i \tag{9}$$

Here β is a design parameter and $\boldsymbol{\theta}_0$ is the "true" model. Set $\boldsymbol{\theta}_0$ as an all-one vector. Then, try Sklearn implementations of SVM and logistic regression and calculate the estimated model $\hat{\boldsymbol{\theta}}_{\text{SVM}}$ and $\hat{\boldsymbol{\theta}}_{\text{LR}}$. There is a Python file "SVM-LR-Risk.py" that helps you write your code. Calculate the square errors $\|\hat{\boldsymbol{\theta}}_{\text{SVM}} - \boldsymbol{\theta}_0\|_2^2$ and $\|\hat{\boldsymbol{\theta}}_{\text{LR}} - \boldsymbol{\theta}_0\|_2^2$. For each value of β, n, d and regularization α, repeat the above 1000 times and average the errors. This gives you a good approximation of the risk value for those parameters. Write a report where,

(a) Plot the value of risk as a function of $\beta \in [.1\ 5]$ for some suitable values of $n \in [100\ 1000]$, $d \in [2\ 20]$ and α.

(b) Discuss what the effect of n, d, α on the risk curves are and discuss why.

(c) Discuss which of the two methods lead to a lower risk curve and why.

Q2. (Bregman divergence) A popular alternative to the the squared 2-norm $\|\mathbf{x} - \mathbf{y}\|_2^2$ in optimization is the Bregman divergence. Given a differentiable, strictly convex function ψ, the Bregman divergence D_ψ is defined as

$$D_\psi(\mathbf{y}, \mathbf{x}) = \psi(\mathbf{y}) - \psi(\mathbf{x}) - \nabla\psi(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \tag{10}$$

Note that $D_\psi(\mathbf{y}, \mathbf{x})$ is non-negative and only attains zero at $\mathbf{y} = \mathbf{x}$.

1. Show that for $\psi(\mathbf{x}) = \|\mathbf{x}\|_2^2$, we have $D_\psi(\mathbf{y}, \mathbf{x}) = \|\mathbf{y} - \mathbf{x}\|^2$ (3p).
2. For the negative entropy function, $\psi(\mathbf{x}) = \sum x_i \ln x_i$ show that $D_\psi(\mathbf{y}, \mathbf{x}) = \sum x_i - \sum y_i + \sum y_i \ln(\frac{y_i}{x_i})$. Note that for probability mass functions ($\sum x_i = \sum y_i = 1$), this expression simplifies to the Kullback Liebler (KL) divergence (5p).